

Executive Summary

This document outlines the business requirements for "SwiftRoute," an intelligent route optimization and delivery management platform for SwiftServe Logistics. The project aims to address the growing inefficiency of current manual routing processes as the company scales its operations. SwiftRoute will dynamically calculate the most efficient delivery routes, provide real-time fleet management capabilities, and offer live updates to customers, thereby significantly improving operational efficiency, enhancing customer satisfaction, and supporting future business expansion.

Project Objectives

- To develop and implement an intelligent route optimization and delivery management platform named "SwiftRoute."
- To significantly improve the efficiency of delivery operations by automating and optimizing routing processes.
- To provide real-time visibility of the entire delivery fleet for effective dispatching and management.
- To enhance customer experience through live tracking and timely delivery notifications.
- To establish a scalable and robust platform architecture capable of supporting future growth and increased delivery volumes.
- To launch a Minimum Viable Product (MVP) within six to eight months.

Needs Statement

SwiftServe Logistics' current manual routing system is becoming increasingly inefficient and unsustainable as the company's operations continue to scale. This inefficiency leads to suboptimal delivery routes, lack of real-time visibility into fleet movements, and an inability to provide timely and accurate updates to customers. There is a critical need for an automated, dynamic, and intelligent system that can optimize delivery routes, manage the fleet effectively, and provide real-time customer communication to maintain operational efficiency and customer satisfaction amidst growth.

Project Scope

In-Scope (MVP): * **Intelligent Route Optimization:** * Dynamic calculation of the most efficient delivery routes. * Ingestion and utilization of real-time data (traffic APIs, weather services, driver reports) for route adjustments. * Consideration of vehicle capacity, delivery time windows, and driver availability in route planning. * **Fleet Management & Dispatch:** * Assignment of drivers to routes. * Real-time vehicle tracking displayed on a map interface. * Capability to re-route drivers mid-journey based on new information or urgent deliveries. * Visibility and control for dispatchers. * **Driver-Facing Mobile Application:** * Reception of assigned routes and navigation instructions. * Functionality to update delivery statuses. * Capability to capture proof of delivery (photos or signatures). * Communication features for drivers to interact with dispatch. * **Customer-Facing Web Portal:** * Provision of a live tracking link for customers. * Display of delivery vehicle's real-time location on a map. * Display of updated Estimated Time of Arrival (ETA). * Display of driver's name and a contact option. * Automated notifications for dispatch, out for delivery, and delivered status. * **Integration:** * Seamless integration with SwiftServe Logistics' existing proprietary Order Management System (OMS) to pull delivery orders and push status updates. * **Architecture:** * Design and implementation of a scalable platform architecture leveraging cloud-native solutions.

Out-of-Scope (for MVP / Later Phase): * Detailed maintenance scheduling within the fleet management aspect. * Development of customer-facing mobile applications (initially, a web-based portal will suffice).

Requirements

Functional Requirements: * **FR1: Route Optimization Engine** * FR1.1: The system shall dynamically calculate the most efficient delivery routes. * FR1.2: The system shall ingest real-time traffic data from external APIs. * FR1.3: The system shall ingest real-time weather data from external services. * FR1.4: The system shall incorporate driver-reported issues (e.g., road closures) into route adjustments. * FR1.5: The system shall factor in vehicle capacity when optimizing routes. * FR1.6: The system shall factor in specified delivery time windows for each order. * FR1.7: The system shall factor in driver availability when assigning routes. * **FR2: Fleet Management & Dispatch Module** * FR2.1: The system shall allow dispatchers to assign drivers to specific routes. * FR2.2: The system shall display the real-time location of every vehicle on a map interface for dispatchers. * FR2.3: The system shall enable dispatchers to re-route drivers mid-journey. * FR2.4: The system shall provide a comprehensive overview of fleet status and location for dispatchers. * **FR3: Driver Mobile Application** * FR3.1: The driver app shall display assigned routes and delivery schedules. * FR3.2: The driver app shall provide turn-by-turn navigation instructions. * FR3.3: The driver app shall allow drivers to update delivery statuses (e.g., "en route," "delivered," "attempted delivery"). * FR3.4: The driver app shall support capturing photo proof of delivery. * FR3.5: The driver app shall support capturing signature proof of delivery. * FR3.6: The driver app shall facilitate communication between drivers and dispatch. * **FR4: Customer Web Portal** * FR4.1: The system shall generate a unique live tracking link for each customer delivery. * FR4.2: The customer portal shall display the delivery vehicle's real-time location on a map. * FR4.3: The customer portal shall display an updated Estimated Time of Arrival (ETA). * FR4.4: The customer portal shall display the assigned driver's name. * FR4.5: The customer portal shall provide a contact option for the driver. * FR4.6: The system shall send notifications to customers for "dispatch" status. * FR4.7: The system shall send notifications to customers for "out for delivery" status. * FR4.8: The system shall send notifications to customers for "delivered" status. * **FR5: Integration with Order Management System (OMS)** * FR5.1: The SwiftRoute system shall seamlessly integrate with SwiftServe Logistics' existing proprietary OMS. * FR5.2: The SwiftRoute system shall pull delivery orders from the OMS. * FR5.3: The SwiftRoute system shall push delivery status updates back to the OMS.

Non-Functional Requirements: * **NFR1: Performance** * NFR1.1: The system shall handle increasing volumes of deliveries without performance degradation. * NFR1.2: The system shall provide real-time updates for vehicle tracking and re-routing with minimal latency. * **NFR2: Scalability** * NFR2.1: The system architecture shall be robust enough to handle significant expansion in fleet size and delivery volume. * NFR2.2: The system shall be designed to leverage cloud-native solutions to ensure scalability. * **NFR3: Usability** * NFR3.1: The driver mobile application shall be intuitive and easy for drivers to use. * NFR3.2: The customer web portal shall provide a clear and user-friendly interface for tracking deliveries. * **NFR4: Reliability** * NFR4.1: The system shall maintain high availability for critical real-time operations (e.g., tracking, re-routing).

Project Constraints

- **Timeline:** The MVP is targeted for launch within six to eight months.
- **Budget:** An initial budget range of ₹50-70 lakhs has been allocated for the initial phase, with some flexibility based on feature justification.
- **Integration:** The project is constrained by the necessity to integrate with SwiftServe Logistics' existing proprietary Order Management System (OMS), for which API documentation will be provided.
- **Technology Stack:** The development team (Innovate Solutions) will leverage cloud-native solutions where appropriate, influencing the architectural design.

Key Stakeholders

Client:

- Ms. Anya Sharma (CEO, SwiftServe Logistics) - Primary business sponsor and decision-maker.
- SwiftServe Logistics Operations Team/Dispatchers - End-users of the fleet management and dispatch module.
- SwiftServe Logistics Drivers - End-users of the mobile application.
- SwiftServe Logistics Customers - End-users of the customer web portal.

Development Partner:

- Mr. Rahul Gupta (Lead Developer, Innovate Solutions) - Responsible for leading the development and technical implementation.
- Innovate Solutions Development Team - Responsible for designing, building, and testing the SwiftRoute platform.

Schedule

- **Proposal Delivery:** Innovate Solutions aims to deliver a detailed proposal to SwiftServe Logistics within the next week (from July 8, 2025).
- **MVP Launch Target:** The target timeline for launching the Minimum Viable Product (MVP) is within six to eight months from project initiation.

Glossary

- **API (Application Programming Interface):** A set of defined rules that enable different software applications to communicate with each other.
- **BRD (Business Requirements Document):** A formal document that outlines the business needs and requirements for a project.
- **Cloud-Native Solutions:** Software applications designed to run in cloud environments, leveraging cloud services for scalability, resilience, and agility.
- **ETA (Estimated Time of Arrival):** The predicted time at which a vehicle or delivery is expected to arrive at its destination.
- **MVP (Minimum Viable Product):** The version of a new product which has just enough features to satisfy early customers and provide feedback for future product development.
- **OMS (Order Management System):** A system used by SwiftServe Logistics to manage customer orders, from placement to fulfillment.
- **Proof of Delivery (POD):** Confirmation that a delivery has been successfully made, often captured through a signature or photograph.
- **SwiftRoute:** The tentative name for the intelligent route optimization and delivery management platform being developed.